Three Novel Deep Learning Solutions for Patch-Wise Underwater Image Quality Assessment

1 Problem Formulation

Underwater images exhibit spatially non-uniform degradations caused by wavelength-dependent light absorption, forward and backward scattering, turbidity, non-uniform illumination, and suspended particles. Consequently, different regions of the same image can exhibit substantially different levels of perceptual quality. Conventional full-image underwater image quality assessment (UIQA) methods generally learn a global representation and directly regress an image-level quality score. Although such approaches can achieve good correlation with human judgments, they do not explicitly model the fact that different spatial regions contribute unequally to perceived image quality.

For example, consider an underwater image containing a fish in the foreground and a large background region consisting primarily of water. A severe degradation affecting the fish may have a much greater influence on human perception than a similar degradation affecting an uninformative background region. Therefore, a desirable UIQA system should distinguish between:

(i) the quality of a local image region;

(ii) the perceptual importance of that region; and

(iii) the degree and type of underwater degradation affecting the region.

Let an underwater image be represented by

$$I \in \mathbb{R}^{H \times W \times 3}. \tag{1}$$

The image is divided into N spatial patches:

$$I \to \mathcal{P} = \{P_1, P_2, \ldots, P_N\}. \tag{2}$$

For every patch P_i, the model estimates a local quality score

$$q_i = f_q(P_i), \tag{3}$$

where

$$q_i \in [0, 1]. \tag{4}$$

A conventional patch-based model can obtain the global quality using uniform average pooling:

$$Q_{\text{avg}} = \frac{1}{N} \sum_{i=1}^{N} q_i. \tag{5}$$

However, this assumes that

$$w_1 = w_2 = \cdots = w_N, \tag{6}$$

which is unlikely to correspond to human visual perception.

Instead, an importance weight w_i is assigned to each patch:

$$Q = \frac{\sum_{i=1}^{N} w_i q_i}{\sum_{i=1}^{N} w_i}. \tag{7}$$

The central research problem is therefore:

$$\boxed{\text{Estimate local quality } q_i \quad \text{and learn perceptual importance } w_i \text{ for every patch.}} \tag{8}$$

2 Solution I: Saliency-Guided Patch Quality Assessment with Adaptive Attention Pooling

2.1 Basic Idea

The first solution introduces a learnable saliency-aware patch pooling mechanism. The central hypothesis is that perceptually important regions should contribute more strongly to the final quality score than visually uninformative regions.

The proposed architecture contains four stages:

$$\boxed{\text{Image} \rightarrow \text{Patch Extraction} \rightarrow \text{Patch Feature Learning} \rightarrow \text{Quality + Saliency} \rightarrow \text{Adaptive Pooling.}} \tag{9}$$

The important distinction from conventional attention pooling is that the attention mechanism is explicitly interpreted as a perceptual importance estimator.

2.2 Patch Extraction

For a patch size $p \times p$, the number of non-overlapping patches is approximately

$$N = \frac{H}{p}\frac{W}{p}. \tag{10}$$

For example, for a 224×224 image with 32×32 patches,

$$N = 7 \times 7 = 49. \tag{11}$$

Thus,

$$\mathcal{P} = \{P_1, P_2, \ldots, P_{49}\}. \tag{12}$$

Overlapping patches can also be used when boundary information and local continuity are important.

2.3 Local Feature Extraction

A CNN backbone extracts a feature representation:

$$\boldsymbol{f}_i = F_\theta(P_i), \tag{13}$$

where

$$\boldsymbol{f}_i \in \mathbb{R}^d. \tag{14}$$

The resulting feature matrix is

$$\boldsymbol{F} = [\boldsymbol{f}_1, \boldsymbol{f}_2, \ldots, \boldsymbol{f}_N]^T \in \mathbb{R}^{N \times d}. \tag{15}$$

Possible backbones include ResNet, EfficientNet, ConvNeXt, or a lightweight CNN designed specifically for underwater imagery.

2.4 Patch-Level Quality Estimation

A quality regression head estimates the quality of each patch:

$$q_i = \sigma\left(\boldsymbol{w}_q^T \boldsymbol{f}_i + b_q\right), \tag{16}$$

where $\sigma(\cdot)$ is a sigmoid function. Hence,

$$q_i \in [0, 1]. \tag{17}$$

2.5 Learnable Saliency Estimation

A second branch estimates perceptual importance:

$$e_i = \boldsymbol{w}_s^T \tanh\left(W_s \boldsymbol{f}_i + \boldsymbol{b}_s\right). \tag{18}$$

The importance weights are obtained using softmax normalization:

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{N} \exp(e_j)}. \tag{19}$$

Therefore,

$$\sum_{i=1}^{N} \alpha_i = 1. \tag{20}$$

The global image quality becomes

$$\boxed{\hat{Q} = \sum_{i=1}^{N} \alpha_i q_i.} \tag{21}$$

Unlike uniform pooling, the proposed model learns which patches deserve greater influence.

2.6 Quality-Aware Saliency

Instead of estimating saliency only from the feature vector, saliency can depend on both the patch representation and its predicted quality:

$$\boldsymbol{z}_i = [\boldsymbol{f}_i; q_i]. \tag{22}$$

The attention energy is then

$$e_i = \boldsymbol{w}_a^T \tanh(W_a \boldsymbol{z}_i + b_a), \tag{23}$$

followed by

$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}. \tag{24}$$

This allows the network to learn interactions between local quality and perceptual importance.

2.7 Training Objective

Most UIQA datasets provide image-level mean opinion scores (MOS), rather than patch-level quality labels. Therefore, weak supervision can be used.

Let the ground-truth MOS be y. The regression loss is

$$\mathcal{L}_{\text{MSE}} = (\hat{Q} - y)^2. \tag{25}$$

For two images I_a and I_b, if

$$y_a > y_b, \tag{26}$$

the predicted scores should satisfy

$$\hat{Q}_a > \hat{Q}_b. \tag{27}$$

A margin-based ranking loss is

$$\mathcal{L}_{\text{rank}} = \max\left(0, m - (\hat{Q}_a - \hat{Q}_b)\right). \tag{28}$$

The total objective is

$$\boxed{\mathcal{L} = \lambda_1 \mathcal{L}_{\text{MSE}} + \lambda_2 \mathcal{L}_{\text{rank}}.} \tag{29}$$

2.8 Novelty

The primary contribution of Solution I is

$$\boxed{\text{Learnable Perceptual Saliency} + \text{Patch-Level Quality} + \text{Adaptive Quality Pooling.}} \tag{30}$$

The model does not assume that every patch contributes equally to human quality perception.

3 Solution II: Degradation- and Saliency-Aware Patch Quality Network

3.1 Motivation

Conventional visual saliency has an important limitation: a visually salient region is not necessarily the region that contributes most to perceived quality degradation. Therefore, the second solution explicitly models underwater degradation.

For every patch,

$$P_i \rightarrow \begin{cases} q_i & \text{local quality,} \\ s_i & \text{saliency,} \\ d_i & \text{degradation.} \end{cases} \tag{31}$$

The final importance becomes

$$w_i = f(q_i, s_i, d_i). \tag{32}$$

3.2 Three-Branch Architecture

The proposed architecture contains three parallel branches:

$$\boxed{\text{Patch} \rightarrow \begin{cases} \text{Quality Branch} \\ \text{Saliency Branch} \\ \text{Degradation Branch} \end{cases} \rightarrow \text{Importance Fusion} \rightarrow \text{Pooling.}} \tag{33}$$

3.3 Quality Branch

The patch feature is

$$\boldsymbol{f}_i = F_q(P_i), \tag{34}$$

and the local quality score is

$$q_i = \sigma(W_q \boldsymbol{f}_i + b_q). \tag{35}$$

3.4 Saliency Branch

The saliency representation is

$$\boldsymbol{s}_i = F_s(P_i). \tag{36}$$

The saliency score is

$$s_i = \sigma(W_s \boldsymbol{s}_i + b_s), \tag{37}$$

where

$$s_i \in [0, 1]. \tag{38}$$

3.5 Underwater Degradation Branch

The degradation representation is

$$\boldsymbol{d}_i = F_d(P_i). \tag{39}$$

The model can estimate multiple degradation attributes:

$$\boldsymbol{r}_i = [r_i^{\text{color}}, r_i^{\text{haze}}, r_i^{\text{blur}}, r_i^{\text{contrast}}, r_i^{\text{noise}}]. \tag{40}$$

A combined degradation score can be defined as

$$d_i = \sum_{k=1}^{K} \beta_k r_i^{(k)}, \tag{41}$$

where

$$\sum_{k=1}^{K} \beta_k = 1. \tag{42}$$

The coefficients β_k can be learned jointly with the network.

3.6 Importance Fusion

The three representations are combined:

$$\boldsymbol{z}_i = [\boldsymbol{f}_i; s_i; \boldsymbol{r}_i]. \tag{43}$$

An importance network estimates

$$e_i = MLP(\boldsymbol{z}_i). \tag{44}$$

The normalized importance is

$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}. \tag{45}$$

The final quality score is

$$\boxed{\hat{Q} = \sum_{i=1}^{N} \alpha_i q_i.} \tag{46}$$

3.7 Multiplicative Importance Formulation

An interpretable alternative is

$$u_i = s_i(1 + \gamma d_i), \tag{47}$$

where $\gamma \geq 0$.

The normalized weight is

$$\alpha_i = \frac{u_i}{\sum_j u_j}. \tag{48}$$

Consequently,

$$\boxed{\hat{Q} = \frac{\sum_i s_i(1 + \gamma d_i) q_i}{\sum_i s_i(1 + \gamma d_i)}.} \tag{49}$$

This formulation increases the influence of a perceptually important region when it is also severely degraded.

3.8 Degradation-Specific Attention

Separate attention maps can be estimated for different degradation types:

$$\alpha_i^{(c)} \tag{50}$$

for color degradation,

$$\alpha_i^{(h)} \tag{51}$$

for haze,

$$\alpha_i^{(b)} \tag{52}$$

for blur, and so on.

The final attention can be calculated as

$$\alpha_i = \sum_{k=1}^{K} \beta_k \alpha_i^{(k)}. \tag{53}$$

This enables the model to learn that different degradation types may affect different regions differently.

3.9 Multi-Task Learning

If degradation labels are available, an auxiliary degradation loss can be introduced. Let the target degradation vector be $\boldsymbol{r}_i^*$. Then

$$\mathcal{L}_{deg} = \frac{1}{N} \sum_i \|\boldsymbol{r}_i - \boldsymbol{r}_i^*\|_2^2 . \tag{54}$$

The overall objective becomes

$$\boxed{\mathcal{L} = \lambda_q \mathcal{L}_{quality} + \lambda_r \mathcal{L}_{rank} + \lambda_d \mathcal{L}_{deg}.} \tag{55}$$

When patch-level degradation annotations are unavailable, pseudo-labels can be generated using synthetic underwater degradation processes.

3.10 Novelty

Solution II explicitly models

$$\boxed{\text{Perceptual Importance} \leftrightarrow \text{Underwater Degradation} \leftrightarrow \text{Quality}.} \tag{56}$$

The model therefore asks not only which region is important, but also how strongly underwater degradation in that region affects the perceived quality.

4 Solution III: Multi-Scale Transformer-Based Patch Quality with Hierarchical Perceptual Pooling

4.1 Motivation

Patch-wise models that process each patch independently may fail to capture contextual relationships between neighboring and semantically related regions. Furthermore, a single patch size cannot simultaneously represent fine-scale distortions and large-scale semantic structures.

Therefore, the third solution introduces

$$\boxed{\text{Multi-Scale Patch Representation} + \text{Vision Transformer} + \text{Hierarchical Importance Pooling.}} \tag{57}$$

4.2 Multi-Scale Patch Extraction

Consider three patch sizes:

$$p_1 = 32, \qquad p_2 = 64, \qquad p_3 = 128. \tag{58}$$

For each scale s,

$$\mathcal{P}^{(s)} = \{P_1^{(s)}, P_2^{(s)}, \ldots, P_{N_s}^{(s)}\}. \tag{59}$$

The feature of a patch at scale s is

$$\boldsymbol{f}_i^{(s)} = F_s(P_i^{(s)}). \tag{60}$$

4.3 Scale-Specific Quality Estimation

For every scale,

$$q_i^{(s)} = Q_s(\boldsymbol{f}_i^{(s)}). \tag{61}$$

The scale-level quality is

$$Q_s = \sum_i \alpha_i^{(s)} q_i^{(s)}. \tag{62}$$

The final image quality can be written as

$$\hat{Q} = \sum_s \lambda_s Q_s, \tag{63}$$

where

$$\sum_s \lambda_s = 1. \tag{64}$$

4.4 Transformer-Based Cross-Patch Interaction

Patch features are converted into a token sequence:

$$\boldsymbol{X} = [\boldsymbol{f}_1, \boldsymbol{f}_2, \ldots, \boldsymbol{f}_N]. \tag{65}$$

The query, key, and value matrices are

$$\boldsymbol{Q} = XW_Q, \qquad \boldsymbol{K} = XW_K, \qquad \boldsymbol{V} = XW_V. \tag{66}$$

The self-attention matrix is

$$A = \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d}}\right). \tag{67}$$

The transformed representation is

$$X' = A\boldsymbol{V}. \tag{68}$$

Thus, the representation of each patch incorporates contextual information from other patches.

4.5 Semantic Importance from Transformer Features

A learnable importance head estimates

$$e_i = MLP(\boldsymbol{h}_i), \tag{69}$$

where $\boldsymbol{h}_i$ is the contextualized Transformer representation.

The normalized importance is

$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)}. \tag{70}$$

The patch contribution is

$$c_i = \alpha_i q_i. \tag{71}$$

4.6 Hierarchical Pooling

Patches can first be grouped into regions:

$$R_k = \{P_i : i \in \mathcal{G}_k\}. \tag{72}$$

The quality of region R_k is

$$Q_{R_k} = \frac{\sum_{i \in \mathcal{G}_k} \alpha_i q_i}{\sum_{i \in \mathcal{G}_k} \alpha_i}. \tag{73}$$

A region-level attention weight is then calculated:

$$\beta_k = \frac{\exp(g(R_k))}{\sum_l \exp(g(R_l))}. \tag{74}$$

The final image quality is

$$\boxed{Q = \sum_k \beta_k Q_{R_k}.} \tag{75}$$

Therefore, the model learns importance at two levels:

$$\boxed{\text{Patch Importance} \rightarrow \text{Region Importance} \rightarrow \text{Image Quality.}} \tag{76}$$

4.7 Cross-Scale Attention

The three scales are represented by

$$F^{(32)}, F^{(64)}, F^{(128)}. \tag{77}$$

Small-patch features can query large-scale semantic features:

$$A_{32 \rightarrow 128} = Softmax\left(\frac{Q_{32} K_{128}^T}{\sqrt{d}}\right). \tag{78}$$

The cross-scale representation becomes

$$H_{32 \rightarrow 128} = A_{32 \rightarrow 128} V_{128}. \tag{79}$$

This allows a fine-scale patch to understand the larger object or structure to which it belongs.

4.8 Scale-Adaptive Quality Fusion

A scale importance score is calculated as

$$e_s = MLP(G_s), \tag{80}$$

where G_s is a global representation of scale s.

The normalized scale weight is

$$\lambda_s = \frac{\exp(e_s)}{\sum_r \exp(e_r)}. \tag{81}$$

The final quality is therefore

$$\boxed{\hat{Q} = \sum_s \lambda_s \sum_i \alpha_i^{(s)} q_i^{(s)}.} \tag{82}$$

4.9 Cross-Scale Consistency Constraint

For a fine-scale patch and the corresponding coarse-scale region, their quality estimates can be encouraged to remain consistent.

Let $M_{ij} = 1$ when fine-scale patch i belongs to coarse-scale region j. A consistency loss is

$$\mathcal{L}_{cons} = \sum_{i,j} M_{ij} \left(q_i^{(32)} - q_j^{(128)} \right)^2. \tag{83}$$

The complete objective becomes

$$\boxed{\mathcal{L} = \lambda_1 \mathcal{L}_{MOS} + \lambda_2 \mathcal{L}_{rank} + \lambda_3 \mathcal{L}_{cons}.} \tag{84}$$

5 Unified Mathematical Formulation

The three solutions can be generalized by defining

$$\boldsymbol{f}_i^{(s)} = F(P_i^{(s)}), \tag{85}$$

$$q_i^{(s)} = Q(\boldsymbol{f}_i^{(s)}), \tag{86}$$

$$s_i^{(s)} = S(\boldsymbol{f}_i^{(s)}), \tag{87}$$

and

$$d_i^{(s)} = D(\boldsymbol{f}_i^{(s)}). \tag{88}$$

The importance energy is

$$e_i^{(s)} = G\left(\boldsymbol{f}_i^{(s)}, q_i^{(s)}, s_i^{(s)}, d_i^{(s)} \right). \tag{89}$$

The normalized importance is

$$\alpha_i^{(s)} = \frac{\exp(e_i^{(s)})}{\sum_j \exp(e_j^{(s)})}. \tag{90}$$

The scale-specific quality becomes

$$Q_s = \sum_i \alpha_i^{(s)} q_i^{(s)}. \tag{91}$$

Finally,

$$\boxed{Q = \sum_s \lambda_s Q_s} \tag{92}$$

or equivalently,

$$\boxed{Q = \sum_s \lambda_s \sum_i \alpha_i^{(s)} q_i^{(s)}.} \tag{93}$$

This equation provides a general mathematical formulation for the proposed patch-wise UIQA framework.

6 Weakly Supervised Multiple-Instance Learning Formulation

Most underwater image quality datasets provide only an image-level MOS. Let the dataset be

$$\mathcal{D} = \{(I_n, y_n)\}_{n=1}^{M}, \tag{94}$$

where y_n denotes the human MOS for image I_n.

The model internally predicts

$$\{q_{n,i}, \alpha_{n,i}\}. \tag{95}$$

The image-level prediction is

$$\hat{y}_n = \sum_i \alpha_{n,i} q_{n,i}. \tag{96}$$

The MOS regression loss is

$$\mathcal{L}_{MOS} = \frac{1}{M} \sum_{n=1}^{M} (\hat{y}_n - y_n)^2. \tag{97}$$

This can be interpreted as a multiple-instance learning problem:

$$\boxed{\text{Image} = \text{Bag}, \qquad \text{Patches} = \text{Instances}.} \tag{98}$$

The image-level MOS supervises the aggregation of latent patch-level quality estimates.

7 Explainability

An important advantage of the framework is that it naturally produces an explanation map. For every patch,

$$(q_i, \alpha_i) \tag{99}$$

are available.

A quality contribution can be defined as

$$c_i = \alpha_i q_i. \tag{100}$$

A degradation contribution can alternatively be defined as

$$r_i = \alpha_i (1 - q_i). \tag{101}$$

A spatial contribution map can then be reconstructed:

$$M(x, y) = r_i, \qquad (x, y) \in P_i. \tag{102}$$

The model can therefore produce:

1. the original underwater image;
2. a patch-quality map;
3. a perceptual-importance map;
4. a degradation map; and
5. the final global quality score.

This provides an explainable UIQA mechanism that identifies regions primarily responsible for perceived quality degradation.

8 Experimental Design

A rigorous experimental study can compare the following models:

$$M_0 = \text{Global CNN}, \tag{103}$$
$$M_1 = \text{Patch CNN + Average Pooling}, \tag{104}$$
$$M_2 = \text{Patch CNN + Attention Pooling}, \tag{105}$$
$$M_3 = \text{Patch CNN + Saliency Pooling}, \tag{106}$$
$$M_4 = \text{Saliency + Degradation Pooling}, \tag{107}$$
$$M_5 = \text{Multi-Scale Patch Model}, \tag{108}$$
$$M_6 = \text{Multi-Scale + Transformer}, \tag{109}$$
$$M_7 = \text{Full Proposed Model}. \tag{110}$$

The purpose is to demonstrate the incremental contribution of every component.

9 Evaluation Metrics

The proposed models should be evaluated using standard UIQA correlation and error metrics.

9.1 Pearson Linear Correlation Coefficient

PLCC measures the linear correlation between predicted and human-assigned quality:

$$PLCC = \frac{\sum_i (\hat{y}_i - \overline{\hat{y}})(y_i - \overline{y})}{\sqrt{\sum_i (\hat{y}_i - \overline{\hat{y}})^2 \sum_i (y_i - \overline{y})^2}}. \tag{111}$$

9.2 Spearman Rank-Order Correlation Coefficient

SROCC evaluates whether the predicted quality ranking agrees with the human ranking. It can be written as

$$SROCC = 1 - \frac{6 \sum_i d_i^2}{N(N^2 - 1)}, \tag{112}$$

where d_i is the rank difference for image i.

9.3 Root Mean Square Error

$$RMSE = \sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}. \tag{113}$$

Kendall's rank correlation coefficient can also be reported.

10 Research Hypotheses

The proposed study can be organized around the following hypotheses.

10.1 Local Quality Hypothesis

Patch-based modeling should outperform global-only modeling:

$$\text{Patch-based UIQA} > \text{Global-only UIQA}. \tag{114}$$

10.2 Importance Hypothesis

Importance-aware pooling should outperform uniform pooling:

$$\text{Importance-aware pooling} > \text{Uniform pooling}. \tag{115}$$

10.3 Degradation Hypothesis

Joint saliency and degradation modeling should outperform saliency alone:

$$\text{Saliency + degradation} > \text{Saliency only}. \tag{116}$$

10.4 Context Hypothesis

Transformer-based contextual modeling should improve over independent patch processing:

$$\text{Context-aware patch modeling} > \text{Independent patch modeling}. \tag{117}$$

10.5 Scale Hypothesis

Multi-scale representation should outperform single-scale representation:

$$\text{Multi-scale UIQA} > \text{Single-scale UIQA}. \tag{118}$$

11 Recommended Research Direction

The three approaches can be treated as progressive stages of a single research program:

$$\boxed{\text{Patch Quality} \rightarrow \text{Saliency-Aware Quality} \rightarrow \text{Degradation-Aware Quality} \rightarrow \text{Multi-Scale Context-Aware Quali}} \tag{119}$$

The most comprehensive model can be summarized as

$$\boxed{\begin{aligned} I &\rightarrow \text{Multi-Scale Patch Extraction} \\ &\rightarrow \text{CNN Feature Extraction} \\ &\rightarrow \text{Transformer Context Modeling} \\ &\rightarrow \begin{cases} \text{Patch Quality} \\ \text{Perceptual Saliency} \\ \text{Underwater Degradation} \end{cases} \\ &\rightarrow \text{Adaptive Importance Estimation} \\ &\rightarrow \text{Hierarchical Quality Pooling} \\ &\rightarrow \text{Global UIQA Score}. \end{aligned}} \tag{120}$$

The central research hypothesis is therefore:

$$\boxed{\text{Global UIQA} = \text{Local Quality} \times \text{Perceptual Importance} \times \text{Underwater Degradation Context}.} \tag{121}$$